

Course Code	Course Title	L	T	P	C
ISWE416L	Big Data Analytics	3	0	0	3
Pre-requisite	NIL	Syllabus version			
		1.0			
Course Objectives					
1. To understand the challenges in Big Data and its analytics methods.					
2. To provide an overview of Apache Hadoop and its Eco System.					
3. To perform real time and batch processing using appropriate algorithms.					
Course Outcomes					
1. Identify big data systems and design for analysis.					
2. Store and analyze data in Hadoop.					
3. Use graph model for solving real world problems.					
4. Process Data using Spark and No SQL Databases.					
5. Apply MapReduce based analysis.					
Module:1	Big Data Concepts	5 hours			
Evolution of Big data, Types and Sources of Data, Characteristics, Analytics Cycle ,Roles in Analytic Projects, Big Data Challenges and Applications in Industries, Different Types of Analytics.					
Module:2	Big Data Platform- Hadoop Storage	5 hours			
Hadoop - History, Terminologies, Distributed File System (DFS), Hadoop Distributed File System (HDFS) - Design, Read and Write in HDFS, Commands - Cluster Architecture- Eco System and Tools.					
Module:3	MapReduce Framework	6 hours			
MapReduce - Different Phases, Shuffle & Sort, Classic - Components - Job Tracker & Task Tracker, Yarn – Components, Workflow – Scheduling - Writing a MapReduce Application.					
Module:4	Real Time Processing	5 hours			
Batch Vs. Real Time Processing –Data Streaming and Analytics - Spark – Architecture, Advantages, Resilient Distributed Dataset (RDD) Operations - Spark on YARN, Functional Programming in Spark, Lambda Architecture - Batch Serving and Stream Layers and Services.					
Module:5	No SQL Database	7 hours			
HBase - Architecture, Create Column Store, Data Definition Language (DDL), Data Manipulation Language (DML) commands, Hive – Architecture, Load Data, Query XML, JSON Files, Cassandra Model - Features, Cassandra Query Language (CQL) - Map, List, Set and Indexes.					
Module:6	Hadoop Tools	7 hours			
Flume Architecture-Reliability and Failure handling; Importing data into HDFS; Importing data into Hive; Pig Architecture, Pig operations, developing and testing Pig Latin scripts, Sqoop.					
Module:7	Big Data Analytical Algorithms	8 hours			
Parallel Frequent Pattern mining - SON, Complementary Naïve Bayes classifier, Random Forest, Decision Tree-based Classifier, Cluster Analysis - Approaches, Parallel K-Means and BFR Algorithm.					
Module:8	Contemporary Issues	2 hours			
Expert Lecture from Industry and R & D Organizations.					
	Total Lecture hours:	45 hours			

Text Book			
1.	DT Editorial Services, “Big Data Black Book”, 1 st Edition, Dreamtech Press, 2017.		
Reference Books			
1.	Jure Leskovec, Anand Rajaraman, Jeffrey David Ullman, “Mining of Massive Datasets”, Cambridge University Press, UK, 2020.		
2.	Bart Baesens, “Analytics in a Big Data World: The Essential Guide to Data Science and its Applications”, Wiley Publishers, 2015.		
3.	Tom White, “Hadoop, the Definitive guide”, O'Reilly Media, 2015.		
4.	David Loshin, “Big Data Analytics: From Strategic Planning to Enterprise Integration with Tools, Techniques, No SQL, and Graph”, Morgan KaufmannElsevier Publishers, 2013.		
Mode of Evaluation: CAT, Written Assignment, Quiz and FAT.			
Recommended by Board of Studies		28-02-2024	
Approved by Academic Council		No. 73	Date 14-03-2024